ORBITS OF $\langle\sigma_4\rangle$ ON PLACES OF SMALL DEGREE OF $X_1(42)_{\mathbb{F}_5}$: MATHEMATICS AND COMPUTATIONAL DETAILS

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Abstract. This note documents the computation of the orbits of the cyclic group $G = \langle\sigma_4\rangle$ of order 12 of automorphisms of $X = X_1(42)_{\mathbb{F}_5}$ on the places of degree $1, \dots, 5$, and explains how the result is stored, labelled independently of the Magma session, and used in the orbit-reduced search of the manuscript "The $\mathbb{Q}$-gonality of $X_1(42)$ is 12" (Section 6 there). Every step is exact. The image of a place is found from the pull-back σ^* alone, with no need to compute the inverse automorphism or any point map. The note is a companion to the scripts `X1_42_orbits.m`, `X1_42_orbits_merge.py`, `run_orbits.sh`, `X1_42_orbits_load.m`, `X1_42_orbits_check.m`, `X1_42_rational_labels.m` and `X1_42_split.py`; it is not part of the manuscript.

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1. Setup

1.1. **The curve.** Let $X = X_1(42)_{\mathbb{F}_5}$; it has genus 25. We use the reduction modulo 5 of Sutherland's plane model $f(x, y) = 0$ (file `FFFc42.txt`); see Lemma 2.5 of the manuscript.

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Magma builds the function field as

`AFF := AlgorithmicFunctionField(FunctionField(XF2))`, $K = \mathbb{F}_5(X) = \mathbb{F}_5(t)[w]/(\Phi(t,w))$, $[K : \mathbb{F}_5$

with $t =$ `BaseField(AFF).1` and $w =$ `AFF.1`. Here $XF2$ is the projective closure of $\bar{f}(x,y) = 0$, and Magma works in the affine chart $y \neq 0$, so that

$$t = \frac{1}{y}, \qquad w = \frac{x}{y}, \qquad \Phi(t,w) = t^{23}\,\bar{f}\Big(\frac{w}{t}, \frac{1}{t}\Big),$$

as in Appendix A.1 of the manuscript (checked by `X1_42_check_tw3.m`, which also writes Φ to `X1_42_Phi.txt`).

Remark 1.1 (Notation in the scripts). The scripts call the generator w "`y`" (variables `YY`, `SL_YY`, `GY`), and t is `TT` or `SL_TT`. Throughout this note we use the notation t, w of the manuscript. The model coordinates x, y do not appear below.

All places are places of $K/\mathbb{F}_5$. For $v \geq 1$ let $\mathcal{P}_v$ be the list `Places(AFF,v)` of places of degree v, in the order in which Magma returns it, and write $P_i^{(v)}$ for its i-th entry. This order is *not* the same in every Magma session (Section 6.3 of the manuscript); Sections 4 and 7 explain how this is handled. The sizes are

v	1	2	3	4	5	6	7	8	9
$\#\mathcal{P}_v$	12	6	8	183	648	2288	11160	48708	214384

The present computation covers $v \leq 5$.

1.2. **The group.** By Proposition A.1 of the manuscript, $\sigma = \sigma_4$ generates a cyclic group G of order 12; the manuscript uses only G (Magma reports that G is the whole of $\mathrm{Aut}_{\mathbb{F}_5}(X)$, but this is not used). The automorphisms are stored in `X1_42_aut_F5.txt` as the pairs $(\sigma_i^* t, \sigma_i^* w) \in K^2$ of pull-backs. The scripts never recompute them: they read the file with `LoadAutDataOpt(name,false)`, which takes about 2 s and skips the closure test. They then check that σ_4 has order exactly 12, by composing σ_4^* with itself until the identity (t, w) appears; this takes about 87 s.

As in Section 4 of the manuscript, Magma returns field automorphisms; for a field automorphism σ we write φ for the automorphism of X with $\varphi^* = \sigma$, and $\sigma(P)$ means $\varphi(P)$.

On the rational places, σ_4 acts as a 12-cycle. In the session-independent labels of Section 7 (file `X1_42_rational_labels.txt`) it is

$$S = [12, 9, 7, 3, 6, 10, 1, 2, 5, 11, 4, 8] = (1\,12\,8\,2\,9\,5\,6\,10\,11\,4\,3\,7). \tag{1}$$

In the numbering of `Places(AFF,1)` in the session of `X1_42_check_orbs.m` it was recorded as

$$A_4 = [9, 5, 2, 1, 12, 8, 11, 7, 3, 4, 10, 6] = (1\,9\,3\,2\,5\,12\,6\,8\,7\,11\,10\,4). \tag{2}$$

Only (1) is used by the search; (2) serves as a consistency check (Section 4).

2. Automorphisms acting on places

2.1. **From the pull-back to places.** Let $\phi \in G$ and write $\phi^* \colon K \to K$ for the corresponding $\mathbb{F}_5$-automorphism of the function field. For a place P with valuation ring $\mathcal{O}_P$, put

$$\mathcal{O}_{\phi(P)} = \{f \in K : \ \phi^* f \in \mathcal{O}_P\} = (\phi^*)^{-1}(\mathcal{O}_P).$$

Since ϕ^* is a field automorphism fixing $\mathbb{F}_5$, this is again a valuation ring of $K/\mathbb{F}_5$. It defines a place $\phi(P)$.

Lemma 2.1. *For every place P and every $f \in K$:*

(a) $v_{\phi(P)}(f) = v_P(\phi^* f)$;

(b) *ϕ^* induces an $\mathbb{F}_5$-isomorphism $\bar{\phi}\colon \kappa(\phi(P)) \to \kappa(P)$ of residue fields, with $\bar{\phi}(f(\phi(P))) = (\phi^* f)(P)$ whenever $f \in \mathcal{O}_{\phi(P)}$. In particular $\deg \phi(P) = \deg P$;*
(c) *$P \mapsto \phi(P)$ is a bijection on the places of each degree v, and $\phi \mapsto (P \mapsto \phi(P))$ is an action of G, possibly through the opposite group, which does not change the orbits.*

Proof. (a) holds because $v_P \circ \phi^*$ is a normalised discrete valuation of K with valuation ring $\mathcal{O}_{\phi(P)}$. For (b), ϕ^* maps $\mathcal{O}_{\phi(P)}$ onto $\mathcal{O}_P$ and, by (a), maps the maximal ideal onto the maximal ideal. It fixes $\mathbb{F}_5$, so the induced map of residue fields is an $\mathbb{F}_5$-isomorphism, and the degree is preserved. For (c), $(\phi^{-1})^*$ provides the inverse map, and $(\phi\psi)^* = \psi^*\phi^*$ shows that composition is respected up to order. □

Because $G = \langle\sigma\rangle$ is cyclic, the G-orbit of a place P is $\{\sigma^k(P) : 0 \le k < 12\}$. Let π_v be the permutation of $\{1, \dots, \#\mathcal{P}_v\}$ with $\sigma(P_i^{(v)}) = P_{\pi_v(i)}^{(v)}$. Then:

Corollary 2.2. *The G-orbits on $\mathcal{P}_v$ are exactly the cycles of π_v. Every cycle length divides 12, and $\pi_v^{12} = \mathrm{id}$.*

So it suffices to compute the single permutation π_v for each degree v. The difficulty is that σ is known only through σ^*. Computing $\sigma(P)$ directly as a place would require the ideal $(\sigma^*)^{-1}(\mathcal{O}_P)$. Instead, we *identify* $\sigma(P)$ inside the already computed list $\mathcal{P}_v$, using only valuations and residues of explicit functions at P. By Lemma 2.1, these are values at $\sigma(P)$ in disguise.

3. Identifying the image of a place

Fix a degree v and a place $P \in \mathcal{P}_v$. The image $\sigma(P)$ is found in at most three exact steps. Each step shrinks a set of candidates that is *guaranteed* to contain $\sigma(P)$.

3.1. **Encoding polynomials.** For a polynomial $g = \sum_{i=0}^{n} c_i x^i \in \mathbb{F}_5[x]$ with $c_n \neq 0$, put

$$\mathrm{Enc}(g) = \sum_{i=0}^{n} c_i 5^i + 5^{n+1} \in \mathbb{Z}_{>0}, \qquad c_i \in \{0, \dots, 4\}.$$

Enc is injective: the leading power 5^{n+1} recovers n, and the base-5 digits below it recover the coefficients. For monic g this is the encoding enc of Section 6.3 of the manuscript. For α in a finite extension of $\mathbb{F}_5$, let $\mathrm{mp}(\alpha) \in \mathbb{F}_5[x]$ denote its minimal polynomial over $\mathbb{F}_5$.

3.2. **Step 1: the bucket.** The place of $\mathbb{F}_5(t)$ under a place Q of K is determined by $\mathrm{mp}(t(Q))$ if t is regular at Q, and is the infinite place if t has a pole at Q. Define the *bucket* of Q as

$$b(Q) = \begin{cases} \mathrm{Enc}(\mathrm{mp}(t(Q))) & v_Q(t) \ge 0, \\ 0 & v_Q(t) < 0. \end{cases}$$

We compute $b(Q)$ for all $Q \in \mathcal{P}_v$ once, and group the indices by bucket.

Lemma 3.1. *Put $b^\sigma(P) = \mathrm{Enc}(\mathrm{mp}((\sigma^* t)(P)))$ if $v_P(\sigma^* t) \ge 0$, and $b^\sigma(P) = 0$ otherwise. Then $b(\sigma(P)) = b^\sigma(P)$.*

Proof. By Lemma 2.1(a), $v_{\sigma(P)}(t) = v_P(\sigma^* t)$, so t has a pole at $\sigma(P)$ iff $\sigma^* t$ has a pole at P. Otherwise $\bar{\sigma}(t(\sigma(P))) = (\sigma^* t)(P)$ by Lemma 2.1(b). An $\mathbb{F}_5$-isomorphism of fields preserves minimal polynomials over $\mathbb{F}_5$. □

So the first candidate set is $C_1 = \{j : b(P_j^{(v)}) = b^\sigma(P)\}$. It contains $\pi_v(i)$ for $P = P_i^{(v)}$, and has at most 16 elements, since $\sum e_i f_i = [K : \mathbb{F}_5(t)] = 16$ over any place of $\mathbb{F}_5(t)$. If $\#C_1 = 1$, the image is found.

3.3. **Step 2: the key.** For a place Q at which t and w are both regular, put $a = t(Q)$ and $c = w(Q)$, and define its *key*

$$\kappa(Q) = \sum_{m=0}^{3} \mathrm{Enc}(g_m)\,(5^{12})^{3-m}, \qquad (g_0, g_1, g_2, g_3) = \big(\mathrm{mp}(a), \mathrm{mp}(c), \mathrm{mp}(a+c), \mathrm{mp}(a+2c)\big).$$

If t or w has a pole at Q, we set $\kappa(Q) = 0$, meaning "not available". All g_m have degree at most $v \le 9$, so $\mathrm{Enc}(g_m) < 5^{11} < 5^{12}$, and κ is an injective function of the quadruple $(g_0, \dots, g_3)$.

Lemma 3.2. *If $\sigma^* t$ and $\sigma^* w$ are regular at P, then $\kappa(\sigma(P))$ equals the key computed from the pair $\big((\sigma^* t)(P), (\sigma^* w)(P)\big)$ in place of (a, c).*

Proof. Use the same isomorphism $\bar{\sigma}$ for both coordinates. It sends $(t(\sigma(P)), w(\sigma(P)))$ to $((\sigma^* t)(P), (\sigma^* w)(P))$, and hence sends $a + c$ and $a + 2c$ to the corresponding combinations. Minimal polynomials over $\mathbb{F}_5$ are preserved. □

The second candidate set is $C_2 = \{j \in C_1 : \ \kappa(P_j^{(v)}) = \kappa^\sigma(P)\}$. It still contains the true image. If $\#C_2 = 1$, the image is found. If C_2 is empty (which can only happen when a key is unavailable), or if $\kappa^\sigma(P)$ cannot be computed, we continue with $C = C_1$; otherwise we continue with $C = C_2$.

Remark 3.3. Two different places can have the same pair (b, κ): for example, two places over the same place of $\mathbb{F}_5(t)$ whose residues (a, c) are conjugate over $\mathbb{F}_5$ only coordinatewise, but not as a pair. This happens for 10 of the 12 rational places. For the places of degree $2 \le v \le 5$ it does not happen: the pairs $(b(Q), \kappa(Q))$ are pairwise distinct in each of these degrees (checked by `X1_42_split.py` and `X1_42_rational_labels.m`; Section 6.3 of the manuscript). Step 3 handles whatever remains.

3.4. **Step 3: the uniformizer pattern.** Let $C = \{j_1, \dots, j_r\}$ with $r \ge 2$, and write $Q_s = P_{j_s}^{(v)}$. For every candidate Q_s, choose a function $u_s \in K$ with $v_{Q_s}(u_s) > 0$. In practice $u_s =$ `LocalUniformizer`(Q_s), and the script checks $v_{Q_s}(u_s) > 0$. Nothing is required of u_s at the other candidates. Record:

- the zero pattern $Z_s = \{q \in \{1, \dots, r\} : \ v_{Q_q}(u_s) > 0\}$, which always contains s;
- the test bit $\beta_s = \big[\, v_P(\sigma^* u_s) > 0 \,\big] \in \{0, 1\}$.

A candidate q *survives* if $q \in Z_s \iff \beta_s = 1$ for every $s = 1, \dots, r$.

Proposition 3.4. *Let q^* be the index with $Q_{q^*} = \sigma(P)$.*

(a) *q^* always survives.*
(b) *If exactly one candidate survives, it is q^*.*
(c) *A candidate $q \ne q^*$ survives only if u_q vanishes at $\sigma(P)$.*

Proof. By Lemma 2.1(a), $\beta_s = [v_{\sigma(P)}(u_s) > 0] = [q^* \in Z_s]$, which proves (a). (b) follows from (a). For (c): $q \in Z_q$, so q can survive only if $\beta_q = 1$, i.e. $v_{\sigma(P)}(u_q) > 0$. □

So Step 3 is never wrong; at worst it is inconclusive. In that case the image is recorded as -1 (*ambiguous*), and the whole degree is rejected at the merge stage (Section 5). In the computation this did not occur: Steps 1–3 determined $\sigma(P)$ for every place of degree ≤ 5.

Example 3.5. Suppose $C = \{1, 2, 3\}$, with zero patterns $Z_1 = \{1\}$, $Z_2 = \{2, 3\}$ and $Z_3 = \{3\}$, and observed bits $(\beta_1, \beta_2, \beta_3) = (0, 1, 1)$. Candidate 1 fails at $s = 1$, since $1 \in Z_1$ but $\beta_1 = 0$. Candidate 2 fails at $s = 3$, since $2 \notin Z_3$ but $\beta_3 = 1$. Candidate 3 satisfies all three conditions. Hence $\sigma(P) = Q_3$. With bits $(0, 1, 0)$ instead, only candidate 2 survives.

3.5. **Cost.** The functions $\sigma^* t$, $\sigma^* w$ and $\sigma^* u_s = u_s(\sigma^* t, \sigma^* w)$ (`ApplyAut`) are exact elements of K. Steps 1 and 2 need two residue evaluations and four minimal polynomials per place. Step 3 needs at most 16 local uniformizers, 16^2 valuations at the candidates, and 16 valuations of pulled-back functions at P. The pulled-back functions are the expensive part, which is why Step 3 is used only when Steps 1–2 leave several candidates. The script reports, for each degree, how many images were decided by each step.

4. Checks

The following are verified by `X1_42_orbits_merge.py` before a degree is saved.

(C1) *Group.* The loaded σ_4 has order exactly 12 (in every job).

(C2) *Consistent numbering.* All parallel jobs for a degree use the same list $\mathcal{P}_v$. Each job writes $\#\mathcal{P}_v$ and the checksum

$$\mathrm{CHK} = \Big(\cdots\big((0 \cdot B + h_1)\,B + h_2\big)\cdots\Big) \bmod M, \qquad h_i = \big(b_i \bmod p\big) + 7\big(\kappa_i \bmod p\big),$$

with $B = 1000003$, $p = 10^9 + 7$, $M = 2^{61} - 1$, and b_i, κ_i the bucket and key of $P_i^{(v)}$. The merge requires all parts to agree.

(C3) *Completeness.* Every index has exactly one image, computed by the job responsible for it, and no image is ambiguous.

(C4) *Permutation.* The images form a permutation of $\{1, \ldots, \#\mathcal{P}_v\}$. By Lemma 2.1(c), this must hold, so it is a non-trivial check of the whole identification.

(C5) *Group law.* All cycle lengths divide 12 (Corollary 2.2).

(C6) *Known case.* For $v = 1$, the permutation equals A_4 in (2) or its inverse. This is a consistency check with `X1_42_check_orbs.m` only: A_4 refers to the numbering of `Places(AFF,1)` in that session, and the inverse is accepted because replacing σ by σ^{-1} replaces π_v by π_v^{-1}, with the same orbits.

If any check fails for a degree, no file is written for that degree, and an old file of that degree is deleted, so a stale result can never be reused.

Two further checks are made later by `X1_42_rational_labels.m`, independently of the session in which the orbit files were made:

(C7) *Same generator and convention in all degrees.* For every place Q of degree $2 \le v \le 5$ (recomputed in the current session), the pair (b, κ) of $\sigma(Q)$ is computed from $\sigma^* t, \sigma^* w$ at Q (Lemmas 3.1 and 3.2) and compared with the pair of the image stored in `X1_42_orbits_d<v>.txt`. Since (b, κ) is unique in these degrees (Remark 3.3), agreement for all places proves that the stored permutations are exactly those of σ_4, with the same convention (σ, not σ^{-1}) as the degree-1 permutation (1). This is recorded in the lines `V v count` of `X1_42_rational_labels.txt`.

(C8) *Rational places.* The permutation (1) on the signature labels is computed directly from the pull-backs and checked to be a 12-cycle.

5. Parallel computation

5.1. **Splitting.** The images of different places are independent, so the work splits trivially. With m parts for degree v, job $k \in \{0, \ldots, m-1\}$ computes $\pi_v(i)$ exactly for

$$i \equiv k + 1 \pmod m.$$

This round-robin split balances the load better than contiguous blocks, because Magma's order of $\mathcal{P}_v$ groups places by the place of $\mathbb{F}_5(t)$ below them, and bucket sizes vary.

Each job still computes the full list $\mathcal{P}_v$ and the buckets and keys of *all* places, since any of them may be a candidate. It also loads the group and checks the order. These shared costs are small compared with the image computations for $v = 4, 5$.

5.2. **Launch and merge.** `run_orbits.sh P1 P2 P3 P4 P5` starts P_v jobs for degree v; the default is $1, 1, 1, 4, 12$, i.e. 19 jobs, one per core:

`magma -b degs:=`v `part:=`k `nparts:=`m `X1_42_orbits.m`

Job (v, k, m) writes `X1_42_orbits_d<v>_part<k>_of_<m>.txt`, which contains:

- a header with `DEG`, `GEN`, `PART`, `NPARTS`, `COUNT` and `CHECKSUM`;
- for part 0 only, the place data lines `P i | c0 c1 ... | key`;
- its lines `I i j`, meaning $\pi_v(i) = j$;
- a closing `END` line.

The log of each job is `log_orbits_d<v>_part<k>_of_<m>.txt`. After all jobs finish, `X1_42_orbits_merge.py` performs checks (C2)–(C6), splits π_v into cycles, and writes the final files.

6. The saved data

For each degree v, the file `X1_42_orbits_d<v>.txt` contains:

line	meaning
`X1_42_ORBITS v1 DEG v GEN 4 COUNT n ORBITS k CHECKSUM c`	header, continued by
`P i | c0 c1 ... cn | key`	place i: coefficients of $\mathrm{mp}(t(P_i))$, constant term first (empty for $t = \infty$), and $\kappa(P_i)$
`I i j`	$\sigma_4(P_i) = P_j$
`O k s : i1 i2 ... is`	orbit k of size s, as the cycle $i_1 \mapsto i_2 \mapsto \cdots$
`R k i`	representative of orbit k (its smallest index)
`END n`	end marker

The indices i are positions in Magma's list $\mathcal{P}_v$ in the session that made the file. For $2 \le v \le 5$ each index is tied to its session-independent data (b, κ) by the `P` line, so the file can be read in any session (Section 7). The `R` lines are not used by the search, which chooses its own representatives (Section 8). `X1_42_orbits_summary.txt` lists, for each degree, the number of orbits and the distribution of orbit sizes.

6.1. **Reuse.** `X1_42_orbits_load.m` provides three functions.

- `LoadOrbitFile(v)` reads the file into a record with fields `coeffs`, `keys`, `img`, `orbits`, `reps` and `orbitof`. It needs no curve computation.
- `PlacesFromOrbitFile(D)` recomputes $\mathcal{P}_v$ and verifies, place by place, the stored polynomial and key and the checksum. If this succeeds, index i in the file refers to `PL[i]` in the current session.
- `FindPlace(coeffs, key, v)` recovers a single place without computing $\mathcal{P}_v$. It takes the zeros of $p(t)$, where p is the stored polynomial (or the poles of t if the data is empty), keeps those of degree v, and keeps the one with matching key. It succeeds when the stored data singles out the place, which is the case for every place of degree $2 \le v \le 5$.

`X1_42_orbits_check.m` runs both reloading routes on every saved degree and prints PASS/FAIL lines. The auxiliary script `X1_42_orbit_reps.m` writes `X1_42_orbit_reps_d<v>.txt` with an explicit description of one place in every orbit; it is not needed by the search.

7. Labels independent of the session

Because Magma's order of $\mathcal{P}_v$ may change between sessions, the search refers to places by labels that depend only on the place.

Degrees $2 \le v \le 5$. The label of a place Q is the rank of $(b(Q), \kappa(Q))$ in sorted order among the places of degree v. This is well defined by Remark 3.3. `X1_42_split.py` translates the permutation π_v of each orbit file into a permutation of these labels, and checks again that it is a permutation with all cycle lengths dividing 12.

Degree 1. Ten of the twelve rational places share their pair (b, κ), so they are labelled instead by *signatures*. `X1_42_rational_labels.m` fixes a list $g_1, \dots, g_n$ of functions: t, w, a local uniformizer at each rational place, and a basis of $\mathcal{L}(3S)$, where S is the sum of the rational places. It writes them to `X1_42_rational_labels.txt` as explicit elements $\sum_{j=0}^{15} c_j(t)\, w^j$ (lines `F i data`), so that every later session reloads exactly the same functions. The signature of a rational place P records, for each g_i, the valuation $v_P(g_i)$ and, if it is 0, the value $g_i(P) \in \mathbb{F}_5$. The twelve signatures are distinct, and the label of P is the rank of its signature in sorted order (lines `R k signature`). The permutation of σ_4 on these labels is computed from the signatures of the pulled-back functions $\sigma^* g_i$, using Lemma 2.1(a),(b), and gives (1) (line `S`). The line `Q1` records the label (5) of the zero of order 3 of w, the place Q_1 of the manuscript; the pole of order 7 of w, the place Q_2, has label 1, and $\sigma_4^5(Q_2) = Q_1$.

8. Use in the divisor search

Let S be the sum of the 12 rational places. Since G permutes the rational places, $\phi(S) = S$ for all $\phi \in G$, and for $\phi \in G$ the map $h \mapsto h \circ \phi^{-1}$ is a degree-preserving bijection $\mathcal{L}(S + E) \to \mathcal{L}(S + \phi(E))$ (Lemma 4.1 of the manuscript).

Every effective divisor E of degree 9 decomposes uniquely as $E = E_{\le 5} + E_{\ge 6}$, with $E_{\le 5}$ supported on places of degree ≤ 5 and $E_{\ge 6}$ either 0 or a single place of degree $6, \dots, 9$ (since $2 \cdot 6 > 9$). By Theorem 6.2 of the manuscript, to prove $\mathrm{gon}_{\mathbb{F}_5}(X) \ge 12$ it suffices to test, for one representative $E_{\le 5}$ of every G-orbit of low-degree parts, all divisors $E_{\le 5} + E_{\ge 6}$, with $E_{\ge 6}$ running over 0 and all places of degree $6, \dots, 9$. The action of σ on places of degree ≥ 6 is never needed.

Write $E_{\le 5}$ as the tuple, over the degrees $v \le 5$ occurring in it in decreasing order, of the sorted lists of labels (Section 7) of its places of degree v, each place listed as often as its multiplicity. Order such tuples lexicographically.

Proposition 8.1. *Keep $E_{\le 5}$ if and only if its tuple is lexicographically smallest among the tuples of $\sigma^k(E_{\le 5})$, $k = 0, \dots, 11$. Then every G-orbit of effective divisors supported on places of degree ≤ 5 contains exactly one kept divisor.*

Proof. σ preserves degrees, so the tuples of $\sigma^k(E_{\le 5})$ have the same shape, and are obtained by applying the stored permutations to the labels and re-sorting. Within an orbit the tuple determines the divisor, so the orbit has a unique lexicographically smallest tuple, and its divisor is the only kept element of the orbit. □

This is the rule defining $\mathcal{R}$ in Section 6.2 of the manuscript, implemented in `kept_low` of `X1_42_split.py`. Since the permutations on the labels are stored, it needs no computation on the curve. As a check, `X1_42_split.py` compares the number of kept parts of each type with the number of G-orbits given by Burnside's lemma from the cycle types of σ alone. The two agree for every partition of 9 (Table 3 of the manuscript). In total 743 231 divisors are tested, compared with 2 994 534 in the first search and 6 370 940 without any reduction. The divisors with $E_{\ge 6} \ne 0$ are split among the jobs by the bucket of the high-degree place, as described in Section 6.3 of the manuscript.

9. Results

The orbits of σ_4 on the places of degree $v \le 5$ (Appendix A.2 of the manuscript) are:

v	$\#\mathcal{P}_v$	orbits	orbit sizes
1	12	1	12
2	6	1	6
3	8	2	$4, 4$
4	183	17	$3,\ 6^2,\ 12^{14}$
5	648	54	12^{54}

All orbit sizes divide 12. In degree 4 the group does not act freely: one orbit consists of places fixed by σ^3, and two of places fixed by σ^6.

Notes on Magma. `LocalUniformizer` is used in Step 3, and the scripts check that it has positive valuation at the place. `UniformizingParameter` and `WeakApproximation([Q],[1])` are not available with these argument types in Magma V2.29-10. Loading the group takes about 2 s, and the order check about 87 s.

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